

Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

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Abstract

001 Evaluating AI/ML systems responsibly requires prac-
002 tices for testing, stress-testing, auditing, comparing,
003 and interpreting specific capabilities—not only aggre-
004 gate benchmark scores. This paper proposes such a
005 practice for spatial reasoning in vision–language mod-
006 els (VLMs): a diagnostic audit of object-relative ori-
007 entation, a capability that aggregate accuracy can hide.
008 On Visual Spatial Reasoning (VSR), moving from a
009 compact 2.2B model to a larger 7B VLM substan-
010 tially improves overall, depth, horizontal, and con-
011 tainment accuracy, but orientation changes only mod-
012 estly; a frontier-scale 2026 model (MiMo-V2.5, 310B
013 total / 15B active) reproduces the pattern (79.5%
014 overall, 65.7% orientation). Except for structured
015 prompting, which degrades performance, trained con-
016 ditions on both models and the frontier-scale zero-
017 shot model remain in a narrow 62–66% orientation
018 band. We test structured prompting, general and tar-
019 geted LoRA, hard negatives, visual/projector adapta-
020 tion, frozen and object-grounded representation probes,
021 and object-centric decomposition; none of the tested
022 configurations reliably closes the gap, and a label-
023 sensitivity analysis (an automated audit with an inde-
024 pendent human re-audit; agreement in the supplemen-
025 tary) shows that the pattern persists after excluding
026 examples flagged as ambiguous.

027 Accuracy also fails to capture relational coherence.
028 LM-side adaptation significantly improves consistency
029 over zero-shot, while hard-negative training raises
030 facing/facing-away consistency from 66.0% to 77.7%
031 with facing accuracy unchanged at 68.9%; the addi-
032 tional pooled consistency gain over LM-only is not sig-
033 nificant. Cross-dataset evaluation on SITE shows that
034 VSR-trained adaptation does not transfer cleanly: it is
035 neutral overall and significantly harms official spatial-
036 relationship reasoning (75.1%→71.6%, $p = 0.004$). A
037 pre-specified orientation-vocabulary slice, frozen before

SITE evaluation, is analyzed as exploratory evidence
rather than as direct replication of the VSR construct.
The results point to a persistent orientation bottleneck
in which orientation is not robustly decodable by the
tested frozen-feature readouts and relational decisions
are not always propagated coherently, while also show-
ing that task-specific spatial adaptation can transfer
poorly. Each component of the audit—the diagnos-
tic ladder, the label-sensitivity check, the accuracy–
coherence separation, and the cross-dataset transfer
protocol—is specified operationally so that it can be ap-
plied by others to other capabilities, benchmarks, and
model families.

1. Introduction

This paper’s contribution is an evaluation practice as
much as a set of findings. We propose a protocol
for testing, stress-testing, auditing, comparing, and in-
terpreting a specific spatial capability across vision–
language models (VLMs), and we demonstrate it on
object-relative orientation, a relation family that ag-
gregate accuracy can hide. Spatial reasoning is central
to multimodal systems that must interact with physical
environments. A useful VLM should not only recognize
objects, but determine whether one object is in front
of another, whether it is facing another, whether two
objects are parallel, and whether equivalent or com-
plementary formulations yield logically compatible an-
swers. These abilities matter for robotics, embodied
agents, navigation, and scene understanding.

Existing work already shows that spatial relations
are difficult for vision–language systems. VSR was
designed as a controlled natural-image probe and
reported particularly poor performance on object-
orientation relations in earlier VLMs [7]. More recent
controlled benchmarks likewise find that strong VLMs
can fail on basic left/right and other grounded rela-
tions [6, 9], while broader evaluations continue to iden-

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075 tify spatial orientation as a difficult factor [15]. Our
076 question is therefore not whether spatial reasoning can
077 fail, but whether the orientation weakness persists in
078 modern generative VLMs of different sizes and after
079 targeted intervention, and what the failure signature
080 reveals about representation, reasoning, and transfer.

081 In our VSR experiments, the distinction between
082 relation families is striking. Moving from SmolVLM2-
083 2.2B-Instruct to the larger Qwen2-VL-7B-Instruct in-
084 creases zero-shot overall accuracy from 74.0% to 80.9%.
085 Horizontal reasoning rises from 70.2% to 84.8%, depth
086 from 68.9% to 75.2%, and containment from 83.4%
087 to 89.3%. Orientation, by contrast, moves only from
088 62.8% to 63.5%. A frontier-scale 2026 model (MiMo-
089 V2.5, 310B total / 15B active parameters) reproduces
090 the same relation-specific pattern: 79.5% overall ac-
091 curacy yet 65.7% on orientation, again inside the
092 62–66% band. We then evaluate interventions span-
093 ning prompting, language-side adaptation, targeted
094 data construction, visual-side adaptation, representa-
095 tion probing, and explicit object-centric decomposition.
096 Structured prompting significantly hurts performance;
097 general and targeted LoRA improve aggregate accuracy
098 but do not remove the orientation ceiling; hard nega-
099 tives improve targeted relational coherence without a
100 significant overall orientation-accuracy gain; and the
101 tested visual-side LoRA configurations do not outper-
102 form the language-side control.

103 As a sensitivity analysis, removing examples flagged
104 by the automated audit raises orientation accuracy
105 (65.7%→78.5% for 7B General LoRA, 66.4%→76.6%
106 for hard-negative LoRA on the strictest 107-example
107 subset) but does not eliminate error (2B structured
108 shows no gain). The exclusion sets are annotator-
109 specific: an independent human re-audit under the
110 same taxonomy reaches only moderate per-item agree-
111 ment ($\kappa=0.36$ on the clean/ambiguous flag) yet repro-
112 duces the same qualitative conclusions; details are in
113 the supplementary. Because the removed examples
114 were drawn from persistent failures, part of this gain is
115 expected by construction.

116 A second failure mode emerges from logical consis-
117 tency. The zero-shot 7B model is consistent on only
118 36.9% of complementary facing/facing-away pairs. LM-
119 side LoRA substantially improves coherence, and hard-
120 negative training raises facing consistency to 77.7%
121 while facing accuracy remains unchanged at 68.9%.
122 The extra hard-negative gain is not significant in the
123 pooled strict-family comparison, yet accuracy stays un-
124 changed while consistency moves: the two metrics cap-
125 ture different behavior.

126 Finally, we test cross-dataset transfer. The 7B
127 model is strong on SITE’s official spatial-relationship-

reasoning category (75.1% raw; 59.2% chance-adjusted
accuracy), yet the VSR-trained adapter does not
generalize cleanly: it is statistically neutral overall
(54.2%→53.8%, $p = 0.66$) and significantly degrades
the official category (75.1%→71.6%, $p = 0.004$). A
pre-specified, non-official orientation-vocabulary slice
is much weaker in the aggregate (48.3% raw; 24.0%
chance-adjusted) and shows no positive transfer, but—
as we detail below—its composition limits what it can
establish, and we therefore do not treat SITE as inde-
pendent confirmation of the VSR orientation construct.

Our contributions are fourfold, each stated as a
reusable evaluation practice rather than as a model-
specific finding:

- A fine-grained diagnostic ladder. Across ten VSR conditions, two differently sized models, and a frontier-scale zero-shot model, several spatial families improve substantially while object-relative orientation remains unusually resistant. The ladder—scaling, prompting, language-side adaptation, targeted data construction, visual-side adaptation, representation probing, and explicit decomposition—is defined operationally so it transfers to other benchmarks and relation families.
- A robustness and sensitivity audit. A label-sensitivity analysis (automated labels with an independent human re-audit), frozen/object-grounded probes, visual-side adaptation, and object-centric decomposition constrain purely annotation-based or single-component explanations. The audit practice (flagged-example exclusion sets, decodability checks against majority baselines) applies to any benchmark whose labels or examples may be ambiguous.
- An accuracy-coherence separation protocol. Complementary-relation tests on VSR’s strict relation pairs show that, for the tested conditions, relational consistency can change substantially while task accuracy remains fixed. The protocol measures consistency against verdict-pattern base rates, so it can be applied wherever complementary formulations of the same relation exist.
- A cross-dataset transfer protocol. SITE shows strong official spatial-relationship performance, while VSR-trained LoRA transfers poorly and significantly harms the official SITE category; the orientation-vocabulary slice is analyzed as exploratory evidence. The protocol (pre-specified subsets, chance-adjusted accuracy, paired transfer tests) is model- and benchmark-agnostic and is reported with negative as well as positive transfer.

None of these practices is specific to orientation or to the two model families studied here; each is specified operationally in the supplementary so that other re-

181 searchers can apply the same audit to other capabili-
182 ties, benchmarks, and models.

183 2. Related Work

184 Spatial reasoning benchmarks. VSR introduced a con-
185 trolled natural-image benchmark of more than 10k
186 image-text pairs spanning 66 spatial relations and ex-
187 plicitly highlighted orientation relations such as fac-
188 ing and parallel as difficult for the VLMs studied at
189 the time [7]. Our work should therefore be read as a
190 modern diagnostic extension rather than the first ob-
191 servation of an orientation weakness. What’sUp iso-
192 lates basic spatial relations by holding object identi-
193 ties fixed while varying configurations, and found large
194 gaps between humans and tested vision-language mod-
195 els [6]. GSR-Bench extends controlled grounded spa-
196 tial evaluation across a larger set of multimodal mod-
197 els and model scales [9]. SITE broadens evaluation
198 to single-image, multi-image, and video inputs and or-
199 ganizes tasks using cognitive-science dimensions includ-
200 ing spatial visualization/orientation, intrinsic/extrinsic
201 relations, and static/dynamic settings; its authors re-
202 port spatial orientation as a persistent challenge [15].
203 Recent broad evaluations such as SpatialScore simi-
204 larly argue that spatial intelligence remains fragmented
205 across tasks and modalities [16], and Mind the Gap
206 isolates spatial reasoning from detection and semantic
207 comprehension in contemporary VLMs [10].

208 Diagnosing spatial failures. Recent work has begun
209 to analyze spatial failures mechanistically rather than
210 only benchmarking them. On VSR itself, AdaptVis
211 traces attention distributions and finds that success-
212 ful spatial reasoning correlates with attention align-
213 ing to actual object locations [2]. CREG charac-
214 terizes the directional organization of attribution evi-
215 dence in Qwen2-VL on spatial-relation benchmarks,
216 showing that standard heatmaps often reflect image
217 layout rather than the queried relation [11]. When
218 More Is Less analyzes when injecting spatial cues, com-
219 mon-sense knowledge, or chain-of-thought instructions
220 helps or hurts VSR, with effects varying by model and
221 setup [1]. Probing Visual Concepts measures linear
222 decodability of visual concepts, including orientation-
223 relevant ones, in lightweight driving VLMs and identi-
224 fies information-flow bottlenecks [13]. VSP decomposes
225 visual spatial planning into perception and reasoning
226 subtasks [17], and MM-Spatial targets 3D spatial un-
227 derstanding with geometric inputs [3].

228 Prior work has thus already established spatial and
229 orientation weaknesses in VLMs and has begun to ana-
230 lyze those failures through prompting, attention, repre-
231 sentation probing, and consistency-based training. Our

contribution is therefore not the first identification of
orientation difficulty or spatial inconsistency. Instead,
we revisit the known VSR orientation weakness in mod-
ern generative VLMs and subject the same relation fam-
ily to a controlled diagnostic ladder spanning adapta-
tion, frozen and object-grounded readouts, visual-side
interventions, complementary-relation consistency, la-
bel sensitivity, and cross-dataset transfer.

Adaptation and spatial specialization. We use LoRA,
which adapts frozen models through trainable low-
rank updates [5], to compare general, targeted, hard-
negative, projector, and vision-plus-projector interven-
tions. This design lets us ask whether benchmark
gains reflect general spatial competence or specializa-
tion. The external SITE transfer experiment is cen-
tral here: VSR adaptation is neutral overall on SITE,
does not improve the orientation subset, and signifi-
cantly degrades SITE’s official spatial-relationship cat-
egory. The result complements benchmark work by
explicitly measuring negative as well as positive cross-
dataset transfer.

Representation probing and logical consistency. Di-
agnostic probes can reveal whether information is ac-
cessible from frozen representations, but probe perfor-
mance must be interpreted cautiously because probe
capacity can itself learn the target task [4]. We there-
fore use linear and nonlinear probes, compare against
majority baselines, add object-grounded features, and
avoid claiming that weak decodability proves informa-
tion is absent. Separately, consistency across trans-
formed or complementary statements has prior litera-
ture, including consistency-oriented training objectives
such as SAGE’s geometric and linguistic duality oper-
ations [8]. Our consistency analysis is a VSR-specific
diagnostic rather than a new consistency framework:
it uses strict complementary relation pairs with ob-
served verdict-pattern and base-rate analysis, and it
shows that instance-level accuracy and complementary-
relation coherence can diverge for the tested conditions.

513 3. Experimental Setup

514 3.1. Benchmarks

VSR. We use VSR [7] as the primary diagnostic
benchmark. The random-split test set contains 2,195
examples. The benchmark defines 64 relation labels;
61 appear in the evaluated test split. The orientation
family contains facing, facing away from, parallel to,
and perpendicular to, totaling 137 test examples. All
main VSR results use the same held-out test set and
deterministic True/False parsing.

SITE. We use SITE [15] for cross-dataset transfer evaluation only and do not train on SITE. The image evaluation contains 2,591 examples: 1,368 single-image and 1,223 multi-image examples. Our primary SITE subset is the official spatial-relationship-reasoning category ($n = 993$). We additionally report a transparently defined, non-official keyword-derived orientation subset ($n = 2,024$), analyzed as exploratory evidence rather than as direct replication of the VSR construct. SITE results use the benchmark’s native multiple-choice format and chance-adjusted accuracy (CAA) in addition to raw accuracy.

3.2. Models and Conditions

Our compact model is SmolVLM2-2.2B-Instruct, and our larger model is Qwen2-VL-7B-Instruct [14]. We evaluate ten canonical VSR conditions: 2B zero-shot, 2B structured prompting, 2B General LoRA, 2B Targeted LoRA, 7B zero-shot, 7B General LoRA, 7B Targeted LoRA, 7B Hard-Neg LoRA, 7B Projector LoRA, and 7B Vision+Projector LoRA. LoRA [5] is used for all parameter-efficient adaptation conditions. We additionally evaluate a frontier-scale zero-shot model, MiMo-V2.5 [12] (310B total / 15B activated parameters, dedicated 729M-parameter vision encoder, 2026), served through its public API with the identical frozen prompt, thinking disabled, and greedy decoding (App. A). Its role is external validation of the diagnosed pattern; it is not fine-tuned.

General LoRA uses a proportional sample across relation families. Targeted LoRA allocates most training examples to weak orientation/depth/horizontal families with rehearsal from the remaining relations. Hard-negative LoRA introduces audited complementary orientation examples, primarily facing/facing-away inversions. Vision-side conditions adapt either the multi-modal projector or upper vision blocks together with the projector. The frozen experimental snapshot records the final result set.

3.3. Metrics and Statistical Tests

For VSR we report accuracy overall and by spatial family. For SITE we report raw accuracy and CAA. Paired condition comparisons use McNemar tests when applicable. Clean-label orientation analysis reports the full orientation set and progressively stricter audited subsets. Logical consistency is evaluated by constructing complementary statements for strict relation pairs (left/right, front/behind, facing/facing-away) and testing whether the model assigns opposite verdicts as required. Parallel/perpendicular is treated separately as a soft complement because both relations can be false for oblique objects.

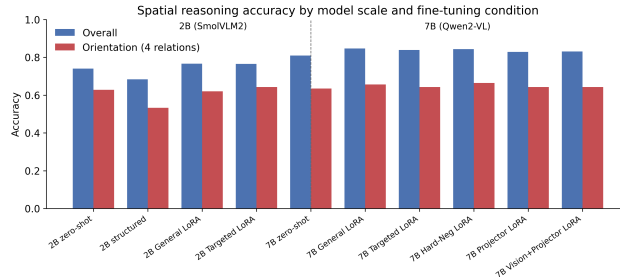


Figure 1. Overall versus orientation performance across the two VLMs and adaptation conditions.

4. Main Results

4.1. Orientation Remains Weak Across Two Differently Sized VLMs

Table 1 shows the central pattern. Moving from the 2B model to the 7B model lifts zero-shot overall accuracy by 6.9 pp, horizontal by 14.6 pp, depth by 6.3 pp, and containment by 5.9 pp, but orientation by only 0.7 pp. General 7B LoRA further raises overall accuracy to 84.7%, yet orientation reaches only 65.7%. Across trained conditions other than structured prompting, orientation remains in a narrow 62–66% band. The frontier-scale zero-shot model (MiMo-V2.5, evaluated identically, App. A) falls in the same band: 79.5% overall and 65.7% orientation, with the orientation family again its weakest relation family (per-family table in the supplementary). The relation-specific pattern is therefore not an artifact of the 2B/7B scale range.

Structured prompting is a negative result: overall accuracy falls from 74.0% to 68.3% at 2B, while orientation falls from 62.8% to 53.3%. A paired exact McNemar test on the full test set confirms that the structured prompt is worse than the minimal baseline ($p \approx 2.2 \times 10^{-7}$). Consistent with recent analyses of information injection on VSR, where additional spatial cues or reasoning instructions can help or hurt depending on the model and setup [1], this result is specific to the tested 2B condition and prompt; it does not establish that prompting generally fails.

4.2. Clean-Label Sensitivity Analysis

An automated audit (LLM-generated labels; the frozen v1 table) identified ambiguous or questionable orientation examples, so we repeat evaluation on progressively stricter subsets as a sensitivity analysis (full table in the supplementary). The best full-set orientation result is 66.4%; on the strictest 107-example subset, the best result rises to 78.5% for 7B General LoRA. Because auditing changes the evaluated sample, these values should not be read as a matched comparison against other re-

Table 1. Canonical VSR test accuracy (%). Orientation improves little with model size or adaptation compared with several other relation families.

Condition	Overall	Orientation	Depth	Horizontal	Containment	Topology/contact
2B zero-shot	74.0	62.8	68.9	70.2	83.4	80.5
2B structured	68.3	53.3	64.9	66.7	78.7	71.7
2B General LoRA	76.6	62.0	71.1	74.3	87.6	81.0
2B Targeted LoRA	76.5	64.2	70.8	75.1	87.6	81.4
7B zero-shot	80.9	63.5	75.2	84.8	89.3	80.3
7B General LoRA	84.7	65.7	82.3	87.4	92.9	84.4
7B Targeted LoRA	83.9	64.2	81.7	87.1	91.7	85.3
7B Hard-Neg LoRA	84.3	66.4	80.7	87.1	89.9	84.6
7B Projector LoRA	82.9	64.2	78.6	87.1	89.3	85.3
7B Vision+Projector LoRA	83.1	64.2	78.0	87.1	89.9	84.4
MiMo-V2.5 zero-shot	79.5	65.7	74.8	81.9	87.1	84.6

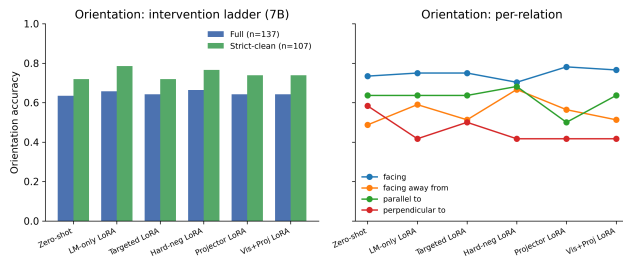


Figure 2. Orientation performance across 7B interventions, including clean-label analysis and per-relation behavior.

Mean-pooled ViT and post-merger features support only weak relation decoding for facing versus facing-away, parallel versus perpendicular, and a four-way orientation task. Linear and MLP probes generally remain near majority baselines; preserving patch-level structure yields a suggestive point estimate for parallel/perpendicular (patch-vote 73.5% vs. 64.7% majority), but that test set is small ($n=34$) and the confidence interval overlaps the majority baseline. In short: orientation is not robustly decodable by the tested frozen-feature readouts.

lation families. A second, independent human re-audit under the same taxonomy (137-example binary flag plus the eight-class taxonomy on the 48 persistent failures) reaches only moderate per-item agreement with the automated labels ($\kappa=0.36$, 66.7% agreement, on the clean/ambiguous flag; Krippendorff’s $\alpha=0.14$ on the taxonomy), yet it reproduces the same qualitative conclusions on the strict 107-example subset: orientation accuracy rises (best 77.6%), error is not eliminated (the human-audit ceiling of 76.6% is below the automated audit’s 78.5%), and 2B structured prompting still shows no gain (51.4%). The exclusion sets are therefore annotator-specific; the frozen table is retained as v1 and the human re-audit table is reported alongside it in the supplementary. The robust conclusion is narrower: removing flagged examples raises orientation scores but does not eliminate error across conditions; the audit is exploratory qualitative evidence, not a ruling on label noise.

5. Why Does Orientation Remain Difficult?

5.1. Frozen Representation Probes

We probe frozen Qwen2-VL-7B-Instruct visual representations without fine-tuning the generative model.

To reduce the confound that global features might infer relations without identifying the relevant objects, we additionally extract subject/reference regions and construct grounded visual features from subject, object, difference, and elementwise-product representations. Object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability. Following standard cautions about diagnostic classifiers [4], we phrase these results as decodability under the tested readouts, not as proof that the information is absent from the network. Linear decodability of visual concepts has been probed in driving-oriented VLMs [13], and directional information in this model family has been inspected via attribution [11]; our probe analysis is not the first of either kind. It differs by targeting VSR’s orientation relation family with pooled, patch-vote, and object-grounded readouts over the exact conditioning features of the generative model, and by pairing decodability with adaptation and consistency interventions rather than treating it as an end in itself. Detailed probe tables and plots are in the supplementary.

5.2. Visual-Side Adaptation and Object-Centric Decomposition

If the frozen representation were the sole limiting factor, directly adapting visual or projector components might recover orientation performance. For the trained checkpoints evaluated here, it did not: projector LoRA and vision+projector LoRA both produced 64.2% orientation accuracy, below the 65.7% LM-only control, while overall accuracy was nominally lower (82.9% and 83.1% vs. 84.7%; unadjusted $p=0.0043/0.0122$; these effects do not survive a blanket multiple-comparison adjustment and are not load-bearing, App. D). This suggests that the tested low-rank adaptation of the visual pathway does not unlock the ceiling.

We also test a box-based two-stage decomposition. Its four-way relation module classifies facing/facing-away/parallel/perpendicular against ground-truth relation labels at 44–46% cross-validation accuracy (49.9% majority baseline; uniform four-class chance 25%), so the tested relation classifier does not outperform the empirical majority baseline. Consistent with that, the pipeline’s True/False verdicts agree with the end-to-end control on only 55.1–58.3% of the 127 boxed orientation statements; in the committed analysis (10 no-box statements scored wrong), verdicts diverge on 61–65 statements, and the control is correct while the pipeline is not in 44–47 of them. The committed pipeline scored agreement with the control rather than ground-truth accuracy, so we report the tested decomposition as failing to reproduce the generative model’s behavior, not as a direct accuracy comparison. Two-dimensional box geometry is structurally insufficient for intrinsic facing direction, and the tested region representations do not supply enough signal to close the gap.

5.3. Logical Consistency Reveals a Distinct Failure Mode

We construct complementary statements for strict relation families. A coherent model should return opposite truth values for left/right, front/behind, and facing/facing-away formulations of the same object pair. Complementary-pair and transformation-based consistency evaluation has prior literature, including consistency-oriented training [8]; our contribution is the VSR-specific diagnostic below, built on strict complementary relation pairs with observed verdict-pattern and base-rate analysis. The zero-shot 7B model is consistent on only 58.0% of left/right pairs, 57.0% of front/behind pairs, and 36.9% of facing pairs. The frontier-scale zero-shot model does not repair coherence either: it reaches 42.7% facing-pair consistency (vs. 36.9% for the 7B zero-shot model), with the same

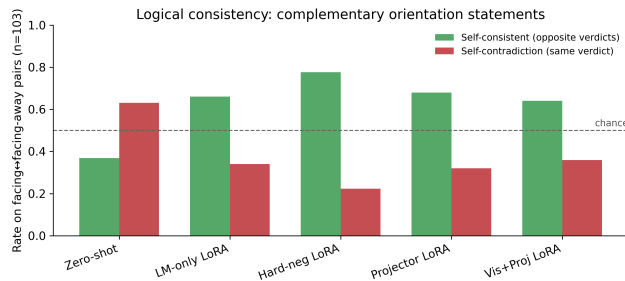


Figure 3. Self-consistency on complementary facing/facing-away relations across 7B conditions.

False-defaulting verdict pattern as the smaller models (0 both-True, 56.0% both-False verdicts; verdict table in the supplementary). Scale alone therefore does not resolve the coherence failure mode. The observed verdict patterns show what drives this: the model affirms either member of a facing pair only 17–20% of the time, a False-defaulting bias that yields both-False verdicts on 63.1% of pairs and no both-True verdicts (0/103); expected consistency under verdict independence given these base rates is 30.2%, below the observed 36.9% (verdict table in the supplementary): the dominant phenomenon is a refusal to affirm either proposition, rather than a preference for contradictory affirmations.

LM-only LoRA raises facing consistency to 66.0% and front/behind consistency to 70.7%; pooled strict-family consistency is significantly better than zero-shot ($p < 0.0001$). Hard-negative training further raises facing consistency to 77.7%. Facing accuracy, however, is identical for LM-only and hard-negative LoRA at 68.9%. The additional hard-negative improvement is not significant in the pooled strict-family McNemar comparison ($p = 0.29$). Paired hard-negative supervision therefore provides a targeted intervention on complementary-relation coherence: it descriptively raises facing-pair consistency while leaving original facing accuracy unchanged, although the additional pooled consistency effect over LM-only is not significant. We do not claim a general coherence algorithm; the key result is separability: identical facing accuracy coexists with substantially different facing consistency. Training also changes the failure mode rather than eliminating it: both-False facing verdicts drop from 63.1% to 9.7% (LM-only) while both-True contradictions rise from 0% to 24.3%, a disclosed specialization trade-off.

6. Cross-Dataset Transfer to SITE

We evaluate the frozen 7B base model and the VSR-trained 7B General LoRA adapter on the same 2,591 SITE image examples using SITE’s native multiple-

Table 2. SITE cross-dataset transfer (images). Entries are raw accuracy / CAA (%).

Subset	Zero-shot	VSR-LoRA	Δ
All images	54.2 / 31.1	53.8 / 30.5	-0.4
Official spatial-rel.	75.1 / 59.2	71.6 / 53.4	-3.5
Orient. heuristic	48.3 / 24.0	48.7 / 24.5	+0.4

choice protocol [15], with no SITE training. Because the pre-specified decision rule resolved to branch 2 (the official category was not broadly weak), this adapter run is a post-protocol cross-dataset transfer check rather than a pre-specified step; the protocol and its outcome are documented in the supplementary. The model is strong on the official spatial-relationship-reasoning category: 75.1% raw accuracy and 59.2% CAA. Applying the VSR-trained adapter does not transfer cleanly: it is statistically neutral overall (54.2%→53.8%, $p = 0.66$) and significantly degrades the official category (75.1%→71.6%, $p = 0.004$; 52 examples fixed vs. 87 broken). VSR gains therefore transfer poorly and can induce negative transfer on the independent benchmark; we do not equate benchmark-specific fine-tuning gains with general spatial-reasoning improvement.

The pre-specified, non-official orientation-vocabulary slice is substantially weaker in the aggregate (48.3% raw / 24.0% CAA vs. 75.1% / 59.2% for the official category, and below the 54.2% / 31.1% all-images average) and shows no positive transfer under the adapter (48.3%→48.7%, $p = 0.70$). Because the slice is keyword-derived and heterogeneous (43% of its examples come from multi-view/cross-image reasoning, only 24% from the spatial-relationship category), we treat it as exploratory evidence rather than as direct replication of the VSR construct. A composition-confound analysis of the 2,591 zero-shot predictions (supplementary §E) supports this caution: after controlling for official category, source dataset, modality, and number of options, the orientation flag is not significantly associated with correctness (odds ratio 0.84, 95% CI [0.60, 1.16], $p = 0.29$), although it remains descriptively 7.5pp lower within the official spatial-relationship category itself (71.3% vs. 78.8%, $n=488/505$). A post-hoc high-precision subset restricted to VSR-construct terms (facing, facing away, parallel, perpendicular; $n=177$) yields 44.1% raw (95% CI [37.0, 51.4]; CAA -2.3%) and is reported as exploratory only. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck; the transfer results stand on the official category, where the model is strong and the adapter demonstrably does harm.



Figure 4. SITE CAA for zero-shot and VSR-trained LoRA. The official spatial-relationship category degrades significantly under transfer.

7. Discussion

Orientation is not spatial reasoning broadly. The strongest conclusion is deliberately narrow. The 7B model is strong on several VSR families and SITE’s official spatial-relationship category. The persistent weakness concerns the object-relative orientation family in VSR; on SITE, the orientation-vocabulary slice is analyzed as exploratory evidence rather than as independent confirmation. This framing is consistent with the original VSR observation that orientation relations are difficult [7], while extending it to modern generative VLMs and a much broader intervention analysis.

Decodability and coherence both matter, as diagnostics. The probes show that orientation is not robustly decodable under the tested frozen-feature readouts, while consistency experiments expose large logical failures even for relations answered above chance. Neither result alone identifies a complete causal mechanism, and both are diagnostic rather than causal-localization claims. Together they support a mixed account in which orientation information is difficult to access reliably under the tested readouts and relational decisions are not always propagated coherently across complementary formulations.

Task adaptation can specialize rather than generalize. General LoRA improves VSR overall accuracy substantially, but its SITE transfer is neutral overall and significantly negative on the official spatial-reasoning subset. Hard negatives improve a targeted coherence behavior while producing relation-specific tradeoffs. Benchmark-specific fine-tuning gains should therefore not be equated automatically with general spatial-reasoning improvement.

8. Limitations

Statistical power is limited at the scale of the orientation diagnostic. The VSR orientation family contains only 137 test examples, and the smallest per-relation subset, perpendicular, has $n=12$; perpendicular results are therefore reported descriptively throughout. The Wilson 95% confidence intervals on the family-level point estimates are correspondingly wide—62.8% [54, 70], 63.5% [55, 71], 65.7% [57, 73], and 66.4% [58, 74] for 2B zero-shot, 7B zero-shot, 7B General LoRA, and 7B Hard-Neg LoRA ($n=137$ each)—so the trained conditions’ 62–66% band should be read as a set of statistically overlapping point estimates rather than a fine-grained ranking. Several smaller point estimates have confidence intervals that include the majority baseline or chance: the patch-vote parallel/perpendicular probe (73.5%, $n=34$; 95% CI [57, 85] vs. the 64.7% majority), the perpendicular subset for the trained 7B conditions (41.7%, $n=12$; [19, 68] vs. 50%), and the 7B zero-shot facing-away subset (48.7%, $n=39$; [34, 64]). Paired McNemar tests on the orientation family rest on small discordant-pair counts and have limited power: the non-significant projector and vision+projector orientation comparisons ($p=0.86$, $p=0.85$) indicate an absence of detectable effect, not evidence of equivalence. Where the paper draws conclusions, it relies on the full-set paired tests and the Holm-adjusted battery in the supplementary rather than on these small- n point estimates alone.

The clean-label audit’s exclusion sets originate from an automated LLM audit: an independent human re-audit reached only moderate per-item agreement ($\kappa=0.36$ on the clean/ambiguous flag; Krippendorff’s $\alpha=0.14$ on the eight-class taxonomy) yet reproduced the qualitative conclusions, so the sets are annotator-specific, and the frozen table is reported as v1 with the human re-audit table alongside. The SITE orientation slice is keyword-derived rather than an official benchmark category; a composition-confound analysis shows its aggregate score is largely explained by task composition, so we do not treat SITE as independent confirmation of the VSR orientation bottleneck. Our probing methods test accessible decodability under specific linear/MLP and pooling choices; failure to decode does not prove information is absent. Visual-side experiments use LoRA rather than full end-to-end re-training. We study two differently sized VLMs from different families plus a frontier-scale third model; this is a cross-model comparison, not a controlled scale experiment. All trained conditions are single-checkpoint runs: the paired McNemar tests quantify disagreement on the fixed test examples, not training-run variability, and these single-seed results do not estimate it. The

MiMo-V2.5 evaluation is served through a public API: 3.1% of its outputs did not conform to the one-word format and are counted as wrong (invalid outputs are never guessed), and 294 responses contained chain-of-thought text alongside the verdict; both are disclosed in the committed predictions (App. A). The SITE evaluation is image-only; SITE video reasoning remains future work. Finally, several interventions are evaluated as negative results, so lack of improvement should be interpreted for the tested configurations rather than as impossibility claims about all prompting, adaptation, or architectural methods.

9. Conclusion

Aggregate spatial accuracy hides a persistent orientation bottleneck in the VLMs studied here. A larger VLM and standard adaptation substantially improve several spatial families while leaving object-relative orientation comparatively resistant, and a frontier-scale 2026 model reproduces the same relation-specific pattern. Auditing, probing, visual-side adaptation, and decomposition constrain simple explanations, while complementary-relation tests show that accuracy and coherence are distinct axes of spatial competence that both remain weak in the frontier model. Cross-dataset evaluation on SITE further shows that VSR-specific adaptation transfers poorly and can negatively transfer to official spatial-relationship reasoning. The results motivate spatial evaluation that reports relation-specific performance, transfer, and logical coherence rather than relying on aggregate accuracy alone.

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